

CRITICAL ANALYSIS OF EXISTING WORK

Project Title

**Explainable Multi-Objective Concrete Mix
Optimization for Strength, Cost, and
Sustainability Using Machine Learning**

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Machine Learning (ML) has emerged as an effective solution for predicting the compressive strength of concrete, reducing the need for time-consuming and expensive laboratory testing. Several studies have applied regression and ensemble learning algorithms such as Random Forest, XGBoost, CatBoost, Support Vector Regression (SVR), Artificial Neural Networks (ANN), and LightGBM to estimate concrete strength using mix design parameters. Most researchers have reported that ensemble learning techniques outperform conventional regression methods due to their ability to capture complex nonlinear relationships between input variables and compressive strength.

Among the reviewed studies, XGBoost consistently demonstrated superior prediction performance because of its boosting mechanism, which minimizes prediction errors and improves generalization. Random Forest also achieved reliable results with reduced overfitting by combining multiple decision trees. CatBoost and LightGBM further enhanced prediction accuracy while reducing computational complexity. Artificial Neural Networks showed excellent performance for large datasets but required extensive training data and parameter tuning.

Although these models achieved high prediction accuracy, most existing research focuses primarily on improving predictive performance while giving limited attention to model interpretability. Many of the proposed models function as "black-box" systems, making it difficult for engineers to understand how individual concrete ingredients contribute to the predicted compressive strength. This lack of transparency reduces confidence in the practical application of machine learning models within the construction industry.

Another limitation observed in the literature is the dependence on laboratory-based datasets. Most datasets are collected under controlled experimental conditions and do not fully represent variations encountered in real-world construction environments. As a result, many developed models may not generalize effectively when applied to practical construction projects involving different materials, environmental conditions, and curing processes.

Furthermore, only a few studies integrate Explainable Artificial Intelligence (XAI) techniques such as SHAP or LIME to interpret prediction results. Similarly, very limited research focuses on optimizing concrete mix proportions while simultaneously predicting compressive strength. Therefore, there remains a need for an explainable, optimized, and practically deployable machine learning framework that provides both accurate predictions and transparent decision support for civil engineers.